CS604P - Quiz No.04

Date: 07 January 2026

Q.1 A system is said to be in a safe state if _____.

A. CPU is idle

B. processes can finish in some order

C. resources are never allocated

D. no process is blocked

Q.2 In Logical to Physical Address Translation, how is the 20-bit Physical Address calculated?

A. (Segment * 16) + Offset

B. Segment * Offset

C. Segment + Offset

D. (Segment / 16) + Offset

Q.3 The Solution of Dining Philosopher Problem is used to avoid _____.

A. Deadlock condition

B. Multitasking condition

C. Isolation condition

D. Separation condition

Q.4 Suppose there are 32 number of frames so number of bits needed for f is:

A. 3

B. 4

C. 2

D. 5

Q.5 In Banker's Algorithm, the system must know the _____ resource demand of each process in advance.

A. minimum

B. current

C. maximum

D. average

Q.6 In Banker algorithm Max data structure is used. If_____, then process Pi can request maximum k instances of resource type Rj.

A. Max[i,k] = j

B. Max[i,j] = k

C. Max[j,i] = k

D. Max[k,j] = i

Q.7 In dining philosopher problem all the five philosophers can do two things _____.

A. sleeping and eating

B. playing and eating

C. laughing and eating

D. thinking and eating

Q.8 Logical Address generated by the CPU is divided into two parts: a page number (p) and a page offset (d) and is calculated by following formula:

A. p / d

B. p + d

C. p * d

D. p % d

Q.9 In dining philosopher problem, a philosopher can start eating once he has _____chopsticks

A. 2

B. 3

C. 5

D. 4

Q.10 For Logical to Physical Address Translation , which register is used to hold the physical address in memory?

A. Stack Pointer (SP)

B. Program Counter (PC)

C. Memory Address Register (MAR)

D. Instruction Register (IR)
